Lecture 17
CONSTRAINT SATISFACTION PROBLEMS

Artificial Intelligence

COSC-3112

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Today's Agenda

- Constraint Satisfaction Problems
- Converting Problems to CSP
- CSP Example: Sudoku
- CSP Example: Map coloring
- Constraint Graph
- Popular Problems of CSP

Constraint Satisfaction Problems (CSP)

- A problem that requires its solution within some **limitations/conditions** also known as **constraints**.
- It consists of:
    - X is a set of variables, {X1, . . . ,Xn}.
    - D is a set of domains, {D1, . . . ,Dn}, one for each variable.
    - C is a set of constraints that specify allowable combinations of values. (C = {C1, C2, C3,......, Cn})

Constraint Satisfaction Problems

- CSP:
  - state is defined by variables X_i with values from domain D_i
  - goal test is a set of constraints specifying allowable combinations of values for subsets of variables

- Allows useful general-purpose algorithms with more power than standard search algorithms

Converting Problems to CSP

- **Step 1:** Create a variable set.
- **Step 2:** Create a domain set.
- **Step 3:** Create a constraint set with variables and domains (if possible) after considering the constraints.
- **Step 4:** Find an optimal solution.

CSP: Example
SUDOKO

Variables ->

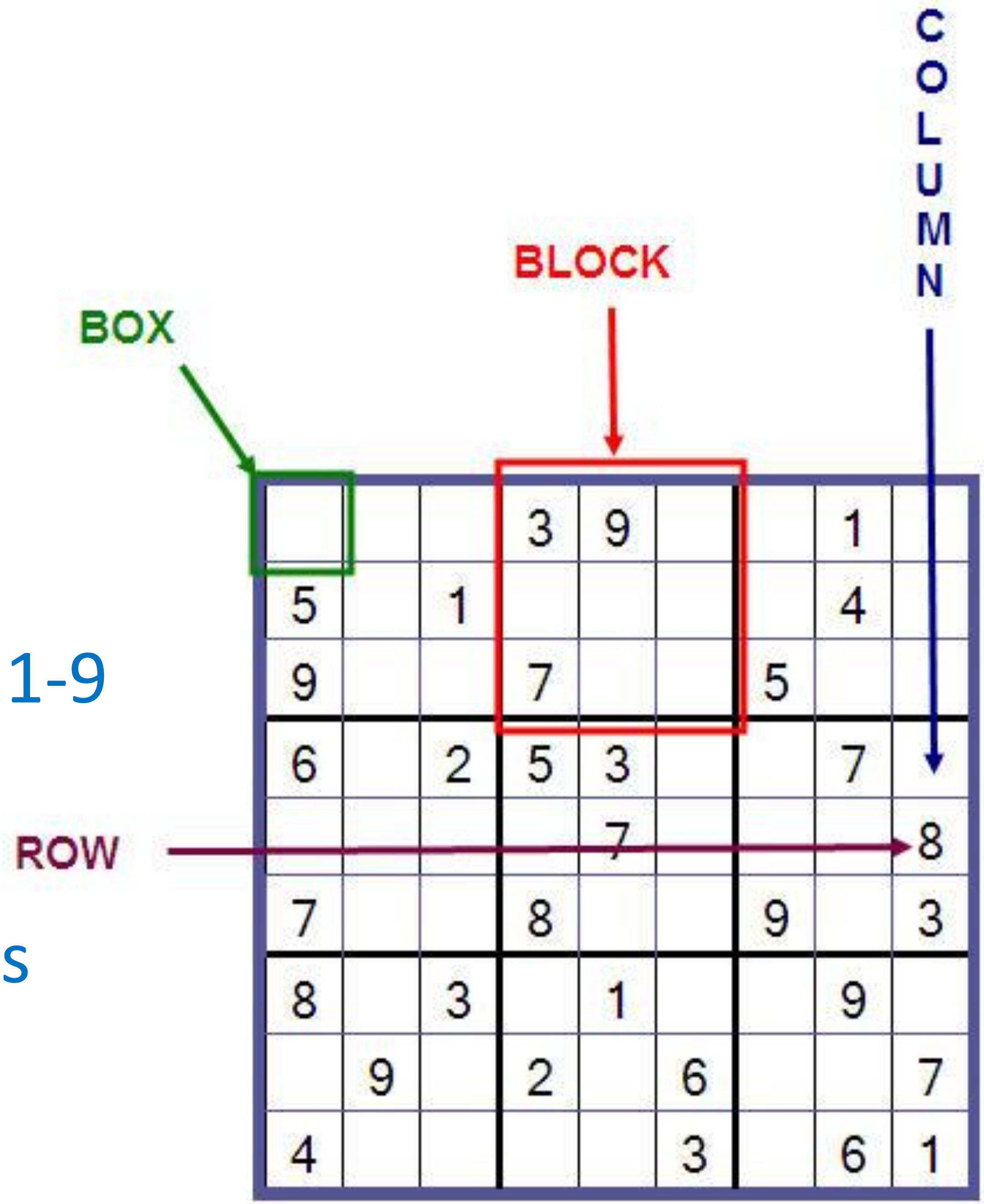
Boxes

Domain ->

Range of Numbers 1-9

Constraints ->

Game Rules



CSP: Example, Map Coloring Problem

- Consider a map coloring problem below:

- A map is given to you and you have to fill it with only three colors : Blue, Green and Red. But no two adjacent locations should have the same color.

CSP: Example Map Coloring

Variables
X={WA, NT, Q, NSW, V, SA, T }

Domains
D_i = {red,green,blue}

Constraints:
adjacent regions must have different colors
e.g., WA ≠ NT

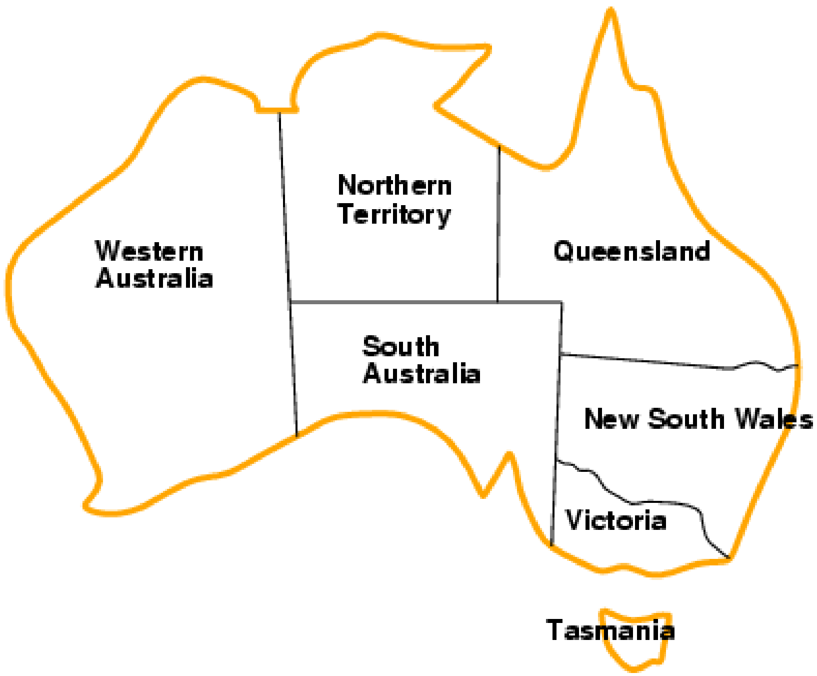
CSP: Example Map Coloring

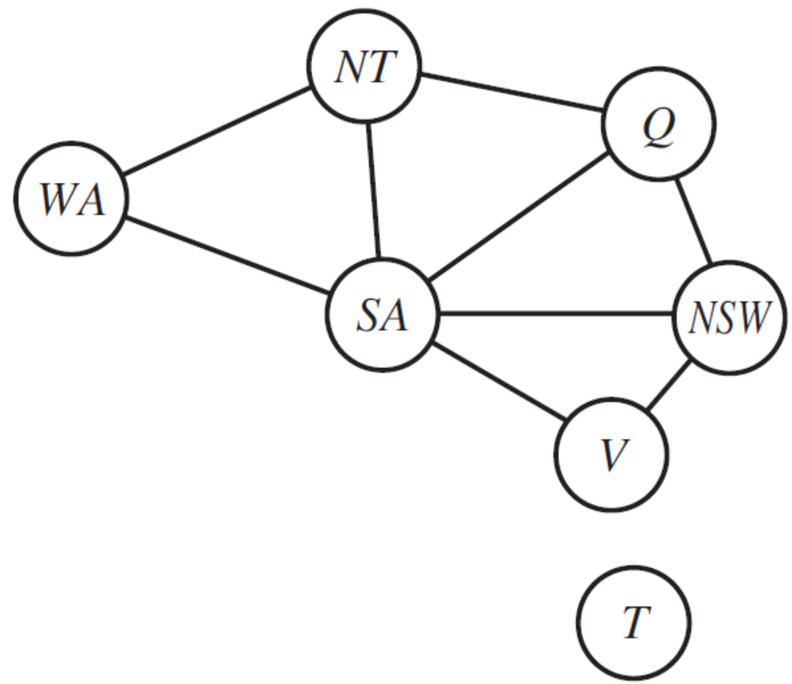
- There are many possible solutions to this problem, such as
  - {WA=red ,NT =green,Q=red ,NSW =green, V =red ,SA=blue, T =red }.

CSP: Constraint Graph

- It is helpful to visualize a CSP as a **constraint graph**
- The nodes of the graph correspond to variables of the problem
- Link connects any two variables that participate in a constraint

CSP: Constraint Graph





CSP: Exercise

CSP: Benefits

- CSPs yield a natural representation for a wide variety of problems
  - it is often easier to solve a problem using CSP based problem solver than to design a custom solution using another search technique
- CSP solvers can be faster than state-space searchers
  - the CSP solver can quickly eliminate large swatches of the search space
- Many problems that are intractable for regular state-space search can be solved quickly when formulated as a CSP

Popular Problems of CSP

- CryptArithmetic (Coding alphabets to numbers)
- n-Queen (In an n-queen problem, n queens should be placed in a nXn matrix such that no queen shares the same row, column or diagonal)
- Map Coloring (Coloring different regions of map ensuring no adjacent regions have the same color)
- Crossword (Everyday puzzles appearing in newspapers)
- Sudoku (A number grid)
- Latin Square Problem

Reading Assignment

- 6.1.2 Example Problem: Job Scheduling
  - from Artificial Intelligence : A Modern Approach 3rd Ed. By S. russel & P. Norvig.